Research Proposal

What Makes AI-Based Sparse-View 3D Gaussian Splatting Work?
Separating the Roles of Predicted Poses, Predicted 3D Points, and Pose Refinement

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1. Summary (abstract)

3D Gaussian Splatting (3DGS) builds a 3D model of a scene from photos, but it needs the camera pose of every photo. With only a few photos, the classic tool COLMAP often fails, so recent methods such as InstantSplat use large pretrained AI models (MASt3R) to predict the camera poses and a dense 3D point cloud, and then refine the poses during training. These methods work well, but it is not clear **which part creates the quality**: the predicted poses, the predicted 3D points, or the refinement. VGGT-X [7] tested two of these parts, but only with hundreds of photos. This project answers that question for only a few photos with a controlled ablation study on the Mip-NeRF 360 dataset using 3, 6 and 9 input views. It uses existing code and pretrained models; it does not propose a new algorithm.

2. Background (simple)

- **3DGS** [1] represents a scene with many small 3D 'blobs' (Gaussians). They are trained with gradient descent so that rendered images match the real photos.
- **Camera poses** tell 3DGS where each photo was taken. Normally they come from **COLMAP**, which needs many overlapping photos.
- **AI foundation models** such as DUSt3R, MASt3R [16] and VGGT [17] are large transformers that predict camera poses and 3D points directly from a few images.
- **InstantSplat** [2] combines MASt3R with 3DGS and corrects the poses during training (joint pose refinement). It reconstructs a scene from 3-12 photos in under a minute.
- **A known evaluation issue**: many sparse-view papers use camera poses computed from ALL photos of the dataset, which is not realistic [5].
- **Closest prior study**: VGGT-X [7] found, with dense views, that pose refinement helps a lot and that random points work about as well as AI-predicted points. Other initialization studies [13, 20, 21] also use many photos or known poses, and SDI-GS [19] uses the same pipeline as InstantSplat without separating its parts.

3. Problem statement

InstantSplat-style methods receive three things from AI: camera poses, dense 3D points, and a refinement step. Published ablations add these parts one by one on other datasets and at a single view count [2], related work tests only one factor at one view count [4], and VGGT-X tests refinement and random points only with dense views [7]. Therefore we do not know how much each part contributes, how far AI poses are from the best possible poses, and whether these answers change when the number of photos changes.

4. Research questions and hypotheses

Main research question: In sparse-view 3D Gaussian Splatting with AI foundation models, how much of the reconstruction quality comes from (1) the predicted camera poses, (2) the predicted 3D points, and (3) joint pose refinement, and how does each contribution change as the number of input views increases from 3 to 9? In particular, do the dense-view findings of VGGT-X [7] still hold with so few photos?

Sub-question	How it is answered	Hypothesis (to be tested, may be wrong)
RQ1: How much quality is lost when AI poses replace the reference poses?	Compare C1 (reference poses) with C2/C3 (AI poses)	The loss is largest with 3 views and smaller with 9 views.
RQ2: How useful are the AI's dense 3D points?	Compare C3 (AI points) with C4 (random points)	With dense views, random points work about as well as AI points [7]; we expect AI points to matter much more with 3 views.
RQ3: When does pose refinement help?	Compare C2 (refinement off) with C3 (on)	With dense views, refinement helps a lot [7]; with few views it may fail when the starting pose error is large [4].
RQ4 (optional): Does a newer AI model help?	Compare C3 (MASt3R) with C5 (VGGT)	VGGT gives better poses, which may improve quality.

5. Method

5.1 Experimental configurations

All configurations use the same InstantSplat code, the same 3DGS renderer, the same training iterations and the same evaluation. Only one part changes at a time.

Config	Camera poses	3D starting points	Pose refinement	Purpose
C1	Reference (from dataset)	MASt3R (aligned to reference)	Off	Upper bound with reference poses
C2	MASt3R	MASt3R	Off	Pure AI prediction, no correction
C3	MASt3R	MASt3R	On	Full InstantSplat (baseline)
C4	MASt3R	Random (same number, uniform in scene box)	On	Value of the AI's 3D points
C5 (optional)	VGGT	VGGT	On	Effect of a newer AI model

5.2 Code changes needed (small, inside InstantSplat)

- **Switch 1 - reference poses:** read the COLMAP poses that come with Mip-NeRF 360 instead of the MASt3R poses.
- **Switch 2 - alignment:** MASt3R points and reference poses use different coordinate systems and scales. Align them with a similarity transform (rotation + translation + scale, Umeyama method) estimated from the camera centres.
- **Switch 3 - random points:** replace the MASt3R point cloud with the same number of random points, sampled uniformly inside a box that encloses the training cameras and the MASt3R points, with a fixed random seed. The exact number of points and the box size are reported, because random-initialization results depend strongly on them [13].
- **Switch 4 - refinement off:** freeze the camera poses during training (learning rate = 0).
- **Switch 5 (optional) - VGGT initializer:** the official repository has an open request for VGGT; a community package of InstantSplat already supports VGGT and can be used as a reference.
- **Pose-error script:** align estimated cameras to reference cameras and compute ATE and RPE.

5.3 Evaluation protocol (follows Free360 [3])

- 8 Mip-NeRF 360 scenes (all except *flowers*, which Free360 excluded because DUSt3R fails on it).
- Test images: the standard 3DGS split (every 8th image is held out for testing).
- Training images: 3, 6 and 9 images sampled evenly from the remaining training images. Camera poses and 3D points are estimated from the training images only; test images are never used for initialization.
- Test-view poses: aligned with the same procedure for every configuration (InstantSplat uses 500 optimization steps with the 3D model frozen), so that all configurations are judged fairly. This test-time alignment is separate from the training-time pose refinement that we switch on and off.
- Image resolution and iterations: the same as the InstantSplat / Free360 code (to be confirmed in week 1).
- Repeatability: 2 scenes are run 3 times with different random seeds to measure noise between runs.

6. Dataset

Mip-NeRF 360 [18]: real indoor and outdoor scenes captured all the way around a central object. It is the standard benchmark for unbounded scenes and is used by 3DGS, FSGS, Free360, SPARS3R and Pi-GS. It already includes camera poses computed by COLMAP from all images, which will be used as the reference poses (very accurate, but not perfect).

Scene	Type	Photos	Used?
bicycle	outdoor	194	Yes
flowers	outdoor	173	No (excluded, as in Free360)
garden	outdoor	185	Yes
stump	outdoor	125	Yes
treehill	outdoor	141	Yes
bonsai	indoor	292	Yes
counter	indoor	240	Yes
kitchen	indoor	279	Yes
room	indoor	311	Yes
Total	5 outdoor, 4 indoor	1,940	8 scenes

- Download: official project page jonbarron.info/mipnerf360. Main file **360_v2.zip (12.5 GB, 7 scenes)**; *flowers* and *treehill* are in a separate 'extra scenes' download on the same page. A Kaggle copy with all 9 scenes also exists.
- Each scene contains the photos, smaller copies of the photos (downsampled 2x, 4x, 8x), and the COLMAP camera files.

7. Metrics

Metric	What it measures	Better when
PSNR	Pixel-level similarity between rendered and real test image	Higher
SSIM	Structural similarity (edges, textures)	Higher
LPIPS	Perceptual similarity using a deep network (closest to human judgement)	Lower
ATE	Absolute camera position error after aligning to the reference cameras	Lower
RPE (translation / rotation)	Error of relative motion between camera pairs	Lower
Training time	Seconds per scene (initialization + training)	Lower
Peak GPU memory	Maximum VRAM used during a run	Lower

Results will be shown as tables (per view count, averaged over scenes), line charts (quality vs number of views for each configuration), and side-by-side rendered images.

8. Technologies and programs

Program / technology	Purpose	Runs on	Status
Windows 11	Operating system of the workstation	-	Installed
NVIDIA driver 591.44	GPU driver	GPU	Installed (workstation)
CUDA Toolkit 12.8 or newer	Compiles the 3DGS GPU code. Required for RTX 5090 (Blackwell)	GPU	To install
Miniconda + Python 3.10/3.11	Separate Python environment for the project	CPU	To install on workstation
PyTorch (CUDA 12.8 build, v2.7 or newer)	Deep learning framework used by all models	GPU	To install
InstantSplat (github.com/NVlabs/InstantSplat)	Main codebase: MASt3R initialization + 3DGS + pose refinement	GPU	To install
MASt3R checkpoint (Naver)	Pretrained AI model that predicts poses and 3D points	GPU	To download
VGGT (facebookresearch/vggt)	Second AI model (optional C5)	GPU	Optional
diff-gaussian-rasterization, simple-knn	CUDA extensions used by 3DGS (compiled during setup)	GPU	Built during setup
COLMAP 4.2.1	Reads the dataset's camera files; optional small COLMAP check	CPU / GPU	Installed
LPIPS, scikit-image / torchmetrics	Image quality metrics	GPU / CPU	To install
NumPy, pandas, matplotlib, Jupyter	Pose-error script, tables and charts	CPU	To install
Git + GitHub	Version control of code and results	CPU	To install
SuperSplat (web) or SIBR viewer	Looking at the 3D results interactively (optional)	GPU	Optional
Overleaf (LaTeX, CVPR template) + Zotero	Writing the paper and managing references	CPU (web)	Online
Claude Code	Assistant for code changes, scripts and documentation	CPU	Available

Plan B if building on Windows fails: use the Windows-compatible fork of InstantSplat, the official Docker image, or WSL2 (Ubuntu inside Windows). This choice will be discussed with the supervisor in week 1.

9. Hardware: what runs on CPU and what runs on GPU

Step	CPU or GPU	Notes
Download and unpack dataset	CPU	Needs disk space (see below)
Select 3/6/9 training views, prepare folders	CPU	Small Python script
Read reference camera poses (COLMAP files)	CPU	Fast
MASt3R / VGGT prediction of poses and points	GPU	Possible on CPU but very slow
Alignment of points to reference poses	CPU or GPU	Small computation
3DGS / InstantSplat training	GPU (required)	CUDA only; cannot run on CPU
Test-view pose alignment and rendering	GPU (required)	Part of InstantSplat evaluation
Image metrics (PSNR, SSIM, LPIPS)	GPU or CPU	LPIPS is faster on GPU
Pose error (ATE, RPE)	CPU	Small script
Tables, charts, writing	CPU	Can be done on the laptop

9.1 Machines

Machine	Specification	Role
Research workstation	NVIDIA GeForce RTX 5090 (32 GB VRAM), Windows 11, driver 591.44, COLMAP 4.2.1	All experiments
Personal laptop	Intel i7-6820HQ, 16 GB RAM, NVIDIA Quadro M1000M (2 GB VRAM), about 6.6 GB free disk	Reading, writing, charts only. Too little VRAM and disk for experiments.

9.2 Estimated resources (estimates, to be confirmed in week 1)

- **Number of runs:** 5 configurations x 3 view counts x 8 scenes = 120 runs.
- **Time per run:** InstantSplat reports about 10-60 seconds per scene on an A100 GPU; with evaluation, a few minutes per run is expected on the RTX 5090. All runs should fit in 1-2 days of computing.
- **GPU memory:** 3-9 images at MASt3R resolution 512 are expected to fit easily in 32 GB.
- **Disk:** dataset about 13-15 GB, software environment about 15-20 GB, results 10-30 GB. At least 100 GB free space is recommended on the workstation.

10. Timeline (4 weeks)

Week	Tasks	Deliverable
1	Install software; download dataset; run InstantSplat on 2 scenes; reproduce Free360 numbers (PSNR about 14.1 / 15.8 / 17.2); learn basics of camera poses	Working setup + reproduction table
2	Implement switches 1-4 and the pose-error script; test all configurations on 1 scene	Tested code
3	Run all experiments (8 scenes x 3 view counts x C1-C5); seed-repeat runs	Full results tables + images
4	Analysis, charts, writing the paper; supervisor feedback	Paper draft + slides

11. Risks and solutions

Risk	Solution
Software does not compile on Windows with the RTX 5090	Use the Windows fork, the Docker image, or WSL2; limit setup to 5 days
Cannot reproduce the Free360 numbers exactly	Report our own reproduced baseline and explain differences (resolution, iterations, view selection)
Alignment of MASt3R points to reference poses is wrong	Check visually in a viewer and with the pose-error script on one scene first
Not enough time	Drop C5 (VGGT); keep C1-C4 and 4 scenes
A very similar paper appears	Repeat the novelty search before submission and adjust the framing

12. Expected contribution and scope

- **Contribution:** a controlled analysis that shows which part of AI-based sparse-view 3DGS matters most (poses, points or refinement), how this changes with the number of photos, and whether dense-view findings [7] still hold with few photos.
- **Outputs:** results tables, comparison images, a short paper (course / thesis chapter / workshop level), and reusable scripts.
- **Type:** analysis / ablation study using existing methods. No new algorithm.
- **Out of scope:** Augmented Reality, Mixed Reality, interactive applications, and new network training.

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